

> LAB 1

↳ 6 cells hidden

> LAB 2

↳ 11 cells hidden

> LAB 3

↳ 4 cells hidden

> LAB 4

↳ 20 cells hidden

✓ LAB 5

```
import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

# XOR dataset
X = np.array([[0,0],
              [0,1],
              [1,0],
              [1,1]], dtype=float)

y = np.array([[0],
              [1],
              [1],
              [0]], dtype=float)

model = keras.Sequential([
    layers.Dense(8, activation='relu', input_shape=(2,)),
    layers.Dense(1, activation='sigmoid')
])
```

```
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

model.fit(X, y, epochs=1000, verbose=0)

loss, accuracy = model.evaluate(X, y, verbose=0)
print("Training Accuracy:", accuracy)

predictions = model.predict(X)

print("\nPredictions:")
for i in range(len(X)):
    print(f"Input: {X[i]} -> Predicted: {predictions[i][0]:.4f}")
```

st-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input\_regularizer=activity\_regularizer, \*\*kwargs`

63ms/step

0.2114
0.7939
0.8744
0.1248

> LAB 6

 ↳ 2 cells hidden

